

Name: kamanisairam

Reg no:192312195

Ex.N.3: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation Loss.

Objective 1:

To compute the Propagation Loss (dB) for Sub-1 GHz, Mid-band, and mmWave frequency ranges using standard path loss models..

Program (Matlab Code)

```
clc; clear; close all;
d = 0.1:0.1:5; % Distance (km)
f = [900 3500 28000]; % Frequency (MHz)
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
figure
% ----- Graph 1 -----
subplot(2,1,1)

plot(d,PL1,'b','LineWidth',2); hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)
title('Propagation Loss vs Distance')
xlabel('Distance (km)')
```



```
ylabel('Loss (dB)')
```

```
grid on
```

```
% ----- Graph 2 (NO legend used to avoid error) -----
```

```
subplot(2,1,2)
```

```
Freq = [900 3500 28000];
```

```
Loss = [PL1(10) PL2(10) PL3(10)];
```

```
plot(Freq, Loss, '-o', 'LineWidth', 2)
```

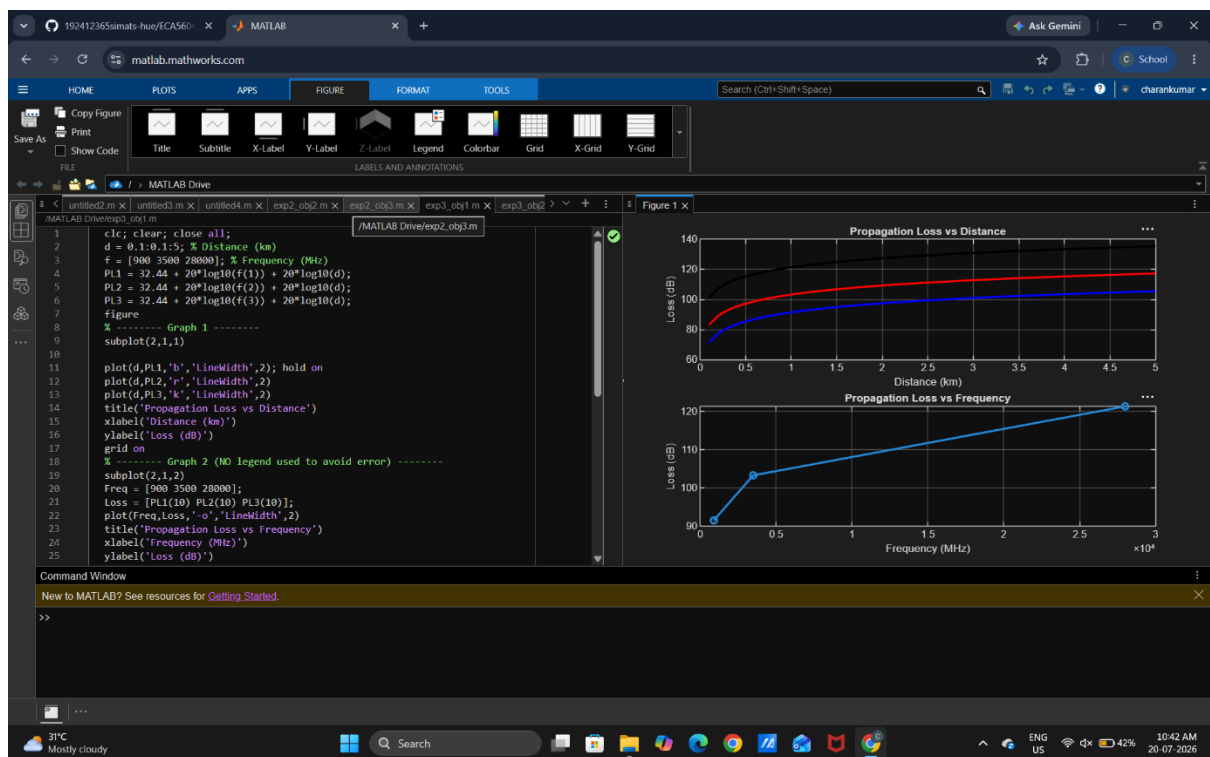
```
title('Propagation Loss vs Frequency')
```

```
xlabel('Frequency (MHz)')
```

```
ylabel('Loss (dB)')
```

```
grid on
```

Output:



Objective 2:

To evaluate the impact of different Operating Frequencies (GHz) on signal attenuation

and coverage performance.

Matlab Code

```
clc; clear; close all;
```

```
d = 2; % fixed distance (km)
```

```
f = [900 3500 28000]; % MHz
```

```
PL = 32.44 + 20*log10(f) + 20*log10(d);
```

```
figure
```

```
% ----- Graph 1: Frequency vs Loss -----
```

```
subplot(2,1,1)
```

```
plot(f,PL,'-o','LineWidth',2)
```

```
title('Propagation Loss vs Operating Frequency')
```

```
xlabel('Frequency (MHz)')
```

```
ylabel('Propagation Loss (dB)')
```

```
grid on
```

```
% ----- Graph 2: Band Comparison -----
```

```
subplot(2,1,2)
```

```
bar(PL)
```

```
title('Comparison of Propagation Loss Across Bands')
```

```
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
```

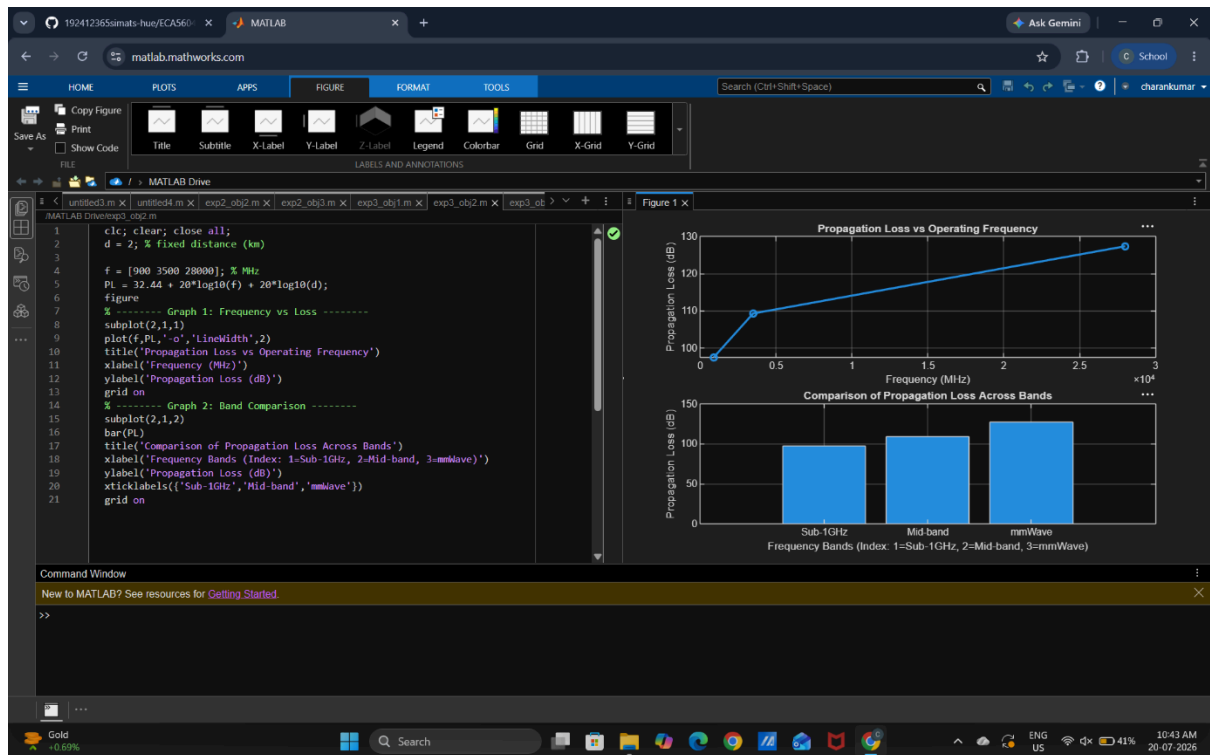
```
ylabel('Propagation Loss (dB)')
```

```
xticklabels({'Sub-1GHz','Mid-band','mmWave'})
```

```
grid on
```

Output





Objective 3:

To analyze how Communication Distance (km) affects propagation loss across different spectrum bands.

Matlab Code

```
clc; clear; close all;
```

```
% Frequency bands (MHz)
```

```
f = [900 3500 28000];
```

```
% Distance (km)
```

```
d = 2;
```

```
% Propagation Loss (FSPL)
```

```
PL = 32.44 + 20*log10(f) + 20*log10(d);
```



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figure

% ----- Graph 1: Simple Line Plot (SAFE) -----

subplot(2,1,1)

plot(f, PL, '-o', 'LineWidth', 2)

title('Propagation Loss vs Operating Frequency')

xlabel('Frequency (MHz)')

ylabel('Propagation Loss (dB)')

grid on

% ----- Graph 2: Safe Comparison Plot -----

subplot(2,1,2)

plot(1:3, PL, '-s', 'LineWidth', 2)

title('Band-wise Propagation Loss Comparison')

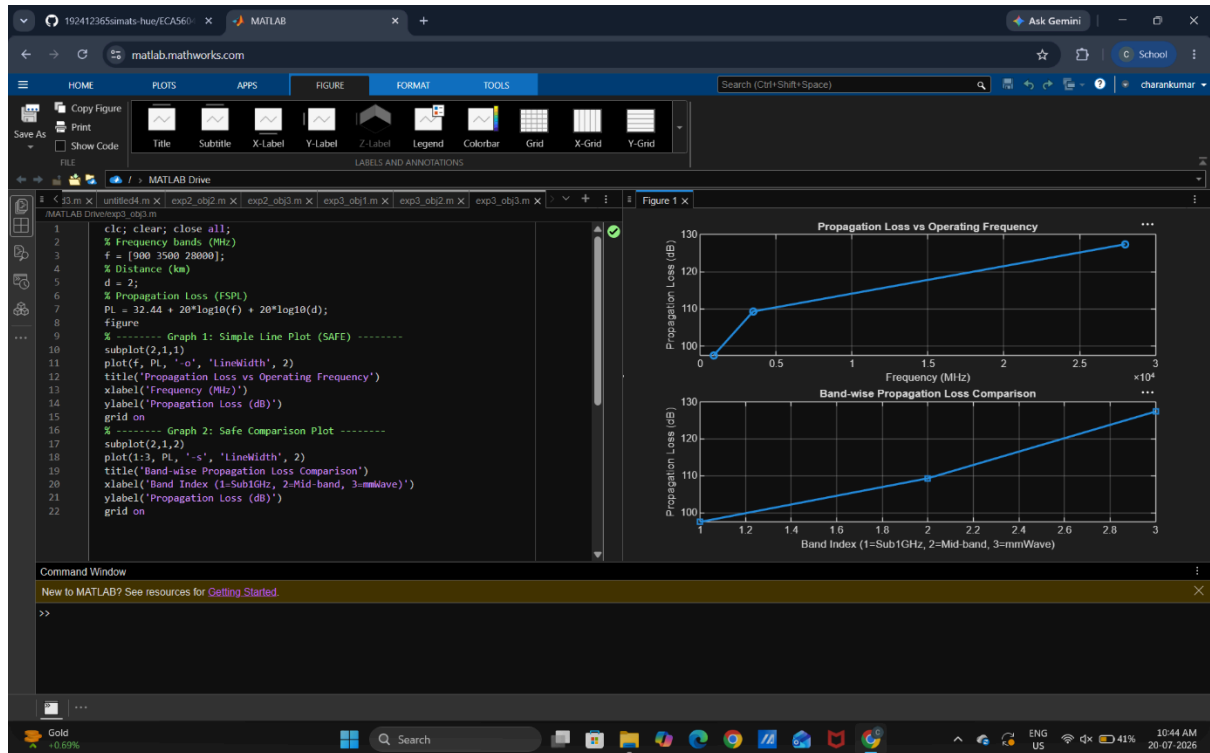
xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')

ylabel('Propagation Loss (dB)')

grid on



Output



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